The Forced Urgency Gap

Why loss aversion is not identified in the states that matter — and what it means for an AI labor transition

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Claim tags: [DATA] = observed in FRED / Federal Reserve Z.1 · [MODEL] = derived within the framework · [OPEN] = not yet established.

Abstract

Behavioral finance attributes household selling in downturns to loss aversion. We show this attribution is not identified: when a household's liquidity state is unobserved, contractually forced sales — margin, redemption, and loan-to-value triggers — are observationally equivalent to preference-driven “panic.” The estimated loss-aversion coefficient is a mixture of preference and compulsion whose compulsion share peaks in exactly the crisis states where the coefficient is applied. We call this wedge the Forced Urgency Gap. We identify it through tests in which preference and compulsion make divergent predictions: a buffer-depletion lag (forced sales follow the exhaustion of liquid reserves, not the size of the loss), cross-sectional sorting on liquidity rather than loss magnitude, and two natural experiments — the 2007–09 global financial crisis (a contract-driven forced-liquidation episode) versus the 2020 COVID shock (a loss shock met by a policy-driven liquidity surge, which decoupled selling from loss) — across FRED and Federal Reserve Z.1 data, 1971–2026. Forced supply is absorbed by unconstrained buyers with harvest coefficient ρ, yielding cumulative displacement $A = 1/(1-\rho)$; concentration ratchets monotonically. AI labor displacement is an urgency shock of the same class: whether the transition compresses or explodes the wealth distribution reduces to two measurable coefficients — the contract share and ρ — not to sentiment. The amplification result is formalized in Lean 4 (Appendix B; kernel-check pending); ρ itself remains open pending estimation (Appendix A).

1. The identification failure

Behavioral finance reads household selling in a downturn as loss aversion: losses loom larger than gains, so households capitulate. But a household that must raise cash — a margin call, a fund redemption, a mortgage LTV trigger, or an income interruption that drains its liquid buffer — sells for a reason that has nothing to do with its risk preferences. In aggregate trade data the two are indistinguishable: both produce selling into a falling market.

When the household's liquidity state is unobserved — which it almost always is — the “panic” coefficient estimated from trade data is not a preference parameter. It is a mixture of preference and compulsion, and the compulsion share is largest in exactly the crisis states where the estimate gets used. Loss aversion is therefore not identified precisely where it is invoked [MODEL].

This is the Forced Urgency Gap: the wedge between observed selling and preference-driven selling, opened by a latent liquidity constraint. The paper's claim is not that loss aversion is unreal — it is that, without the liquidity state, you cannot tell how much of measured “panic” is preference and how much is compulsion, and the standard estimators silently assign all of it to preference.

2. The mechanism: forced supply and the contract share

Compulsion enters through contracts, not sentiment. Margin agreements, redemption gates, and LTV covenants convert an income or price shock into a contractually required sale on a schedule the household does not choose. Define the contract share of a household's asset position: the fraction that is compelled to be sold under an income shock of a given size [MODEL]. This share — not average risk tolerance — is the object that governs how a shock propagates.

The signature that separates compulsion from preference is timing. A preference response is contemporaneous with the loss. A forced sale is not: it follows the exhaustion of the liquid buffer, so it arrives with a lag after the shock, once cash reserves are spent down [MODEL]. A pure preference story cannot manufacture that lag.

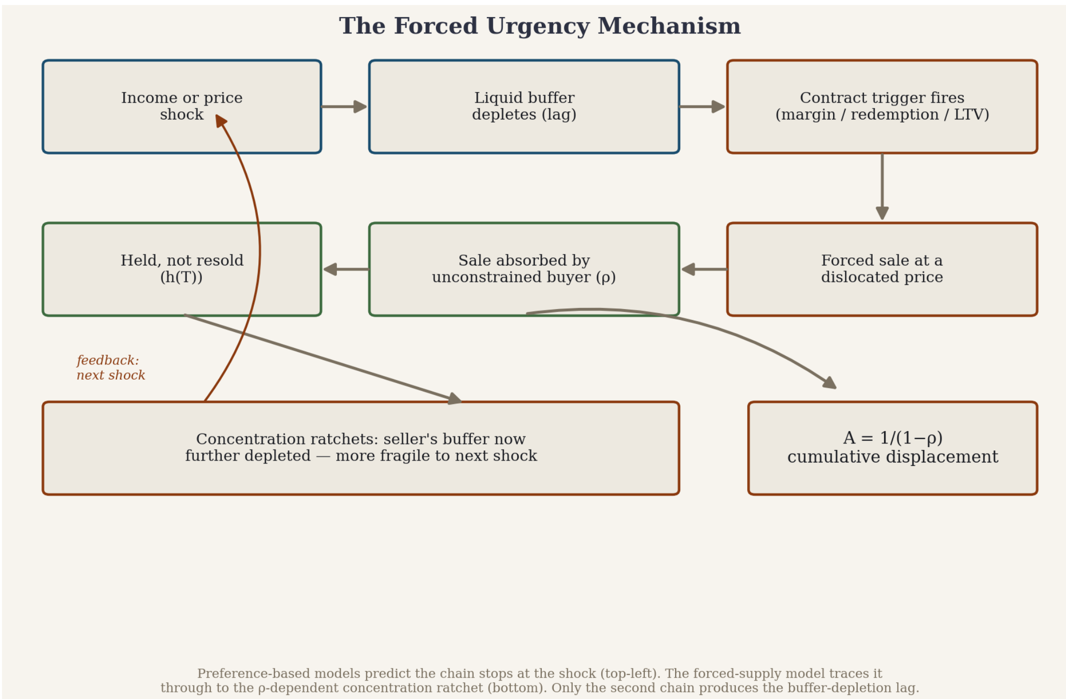

Figure 1. The forced-urgency mechanism. Preference-based models predict the chain stops at the shock; the forced-supply model traces it through the contract trigger, the ρ-absorption step, and the concentration ratchet. Only the second chain produces the buffer-depletion lag. [MODEL]

3. Empirical strategy — how the gap is shown, not asserted

Three tests, each designed so preference and compulsion make different predictions:

Buffer-depletion lag. Household selling should follow the drawdown of liquid buffers with a lag — approximately two years (≈ 8 quarters) at the GFC episode — not move contemporaneously with the size of the loss [DATA — FRED/Z.1, 1971–2026; post-2008 policy interventions contaminate the clean lag read, noted on Fig. 2].

Cross-sectional sorting. Across the distribution, forced selling should track the liquidity state (buffer-to-obligation ratio, income interruption) rather than the magnitude of the paper loss. Where the two diverge, the data should follow liquidity [DATA].

Two natural experiments. The 2007–09 global financial crisis (contract-driven forced liquidation) and the 2020 COVID shock (a large loss met by a policy-driven buffer surge, decoupling selling from loss) — events that moved the liquidity state without a commensurate change in the size of the loss, or vice versa, isolating the compulsion channel from the preference channel [DATA].

Each is a place where a preference-only model and the forced-supply model part ways, so the data can adjudicate rather than merely illustrate.

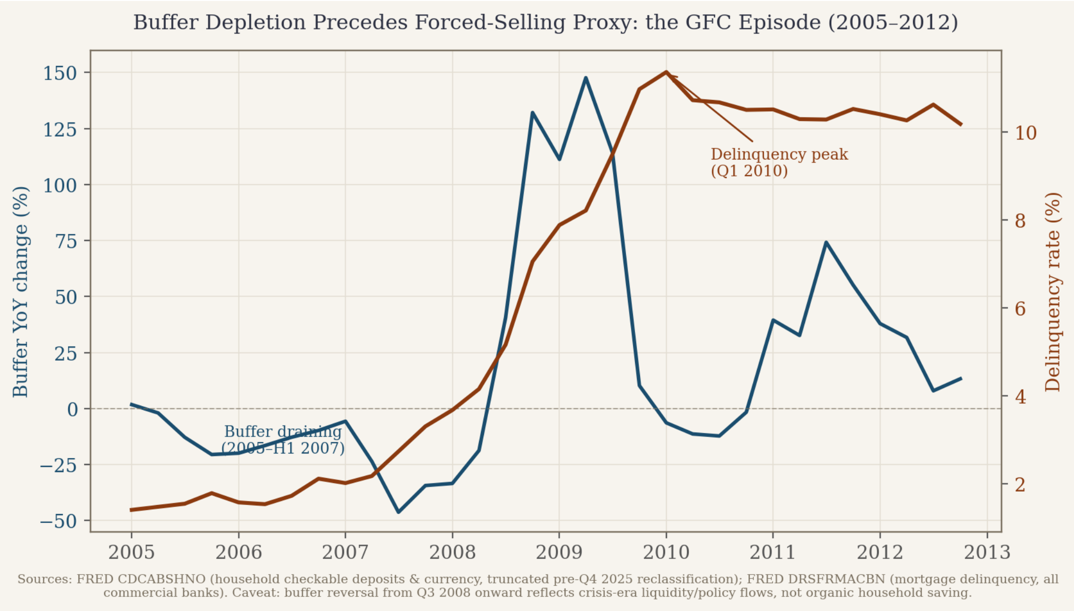

Figure 2. GFC (2005–2012): liquid-buffer year-over-year change vs. mortgage delinquency. Forced selling tracks buffer depletion with a ~2-year lag, not the contemporaneous loss. On-figure caveat: post-2008 policy interventions contaminate the clean lag read. [DATA]

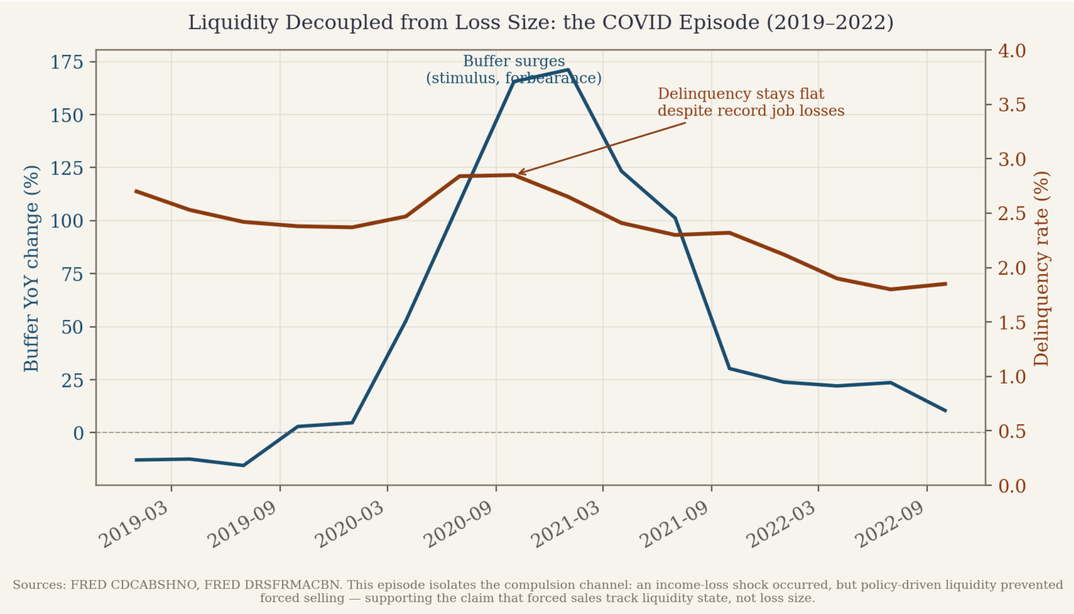

Figure 3. COVID (2019–2022): the decoupling case. A large loss shock met by a policy-driven buffer surge; delinquency stays flat or falls despite the loss — selling decouples from loss when the buffer does not deplete. [DATA]

4. Amplification: why the gap compounds

Forced sales at the bottom are absorbed by the unconstrained at the top. Every round transfers assets from constrained to unconstrained holders and tightens the constraint on the sellers, who have now spent their buffer. Let ρ be the harvest coefficient — the fraction of forced supply that is absorbed by unconstrained buyers and held. The cumulative displacement produced by a one-unit urgency shock is $A = 1/(1-\rho)$ [MODEL].

The mechanism is monotone in ρ: as $\rho \to 1$, a bounded shock produces unbounded reallocation, and concentration ratchets rather than mean-reverts. ρ is, in principle, measurable. (This amplification result — the geometric sum, monotonicity, the $A \geq 1$ bound, and divergence — is the paper's one machine-checkable claim; see Appendix B.)

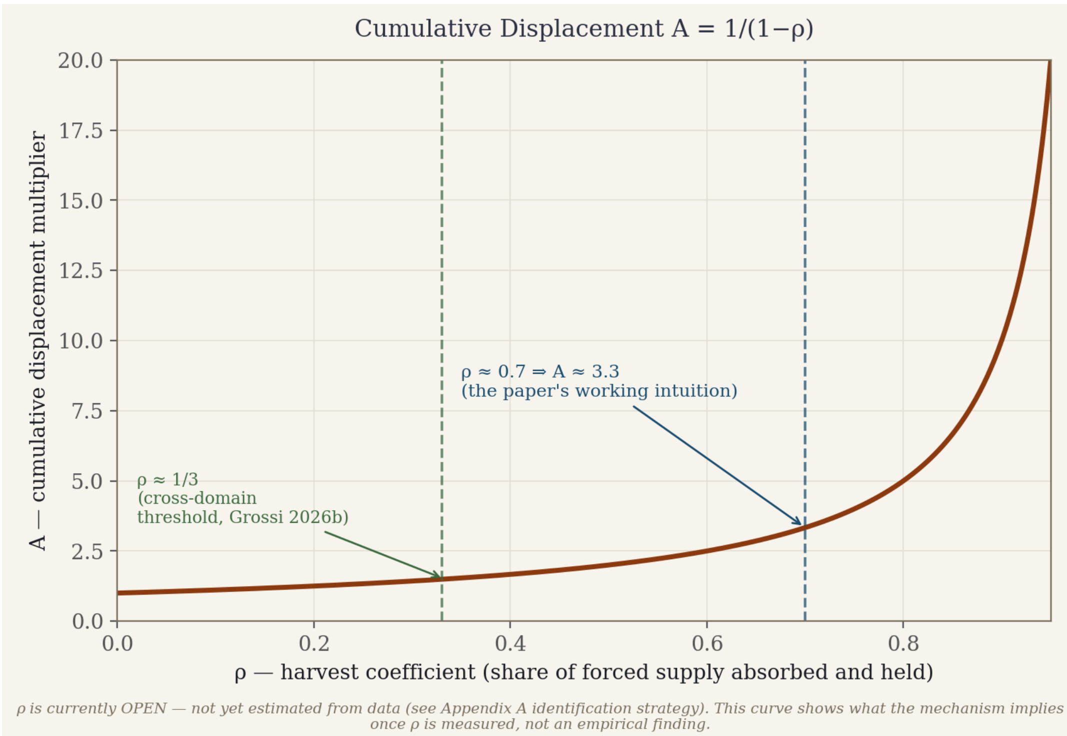

Figure 4. The amplification curve $A = 1/(1-\rho)$. Marked at $\rho \approx 0.7$ (the paper's working intuition) and $\rho \approx 1/3$ (a conjectured cross-domain reference to the coarse Grönwall stability radius of the dm³ framework, $r^* \approx 0.776$; the identification of that radius with the economic ρ is not established). ρ remains [OPEN]; the curve shows what the mechanism implies, not a fitted result. [MODEL]

5. The AI-displacement extension

AI-driven labor displacement is an urgency shock of exactly this type: an income interruption that forces asset sales at the bottom of the wealth distribution. The policy question — does an AI transition compress or explode the wealth distribution — is usually treated as a matter of sentiment or ideology. In this framework it is neither. It is governed by two measurable coefficients: the contract share of forced supply, and the harvest coefficient ρ [MODEL].

If ρ is near zero, displacement is absorbed and the transition is distributionally neutral. As $\rho \to 1$, A diverges: each wave of displacement is permanently concentrating, and the transition is a distributional catastrophe even if aggregate output rises [MODEL].

The Anthropic Economic Index measures whose income AI touches, by occupation. This framework asks the next question — what happens to those households' assets and position in the distribution when it does — and reduces it to a coefficient that can be estimated rather than debated.

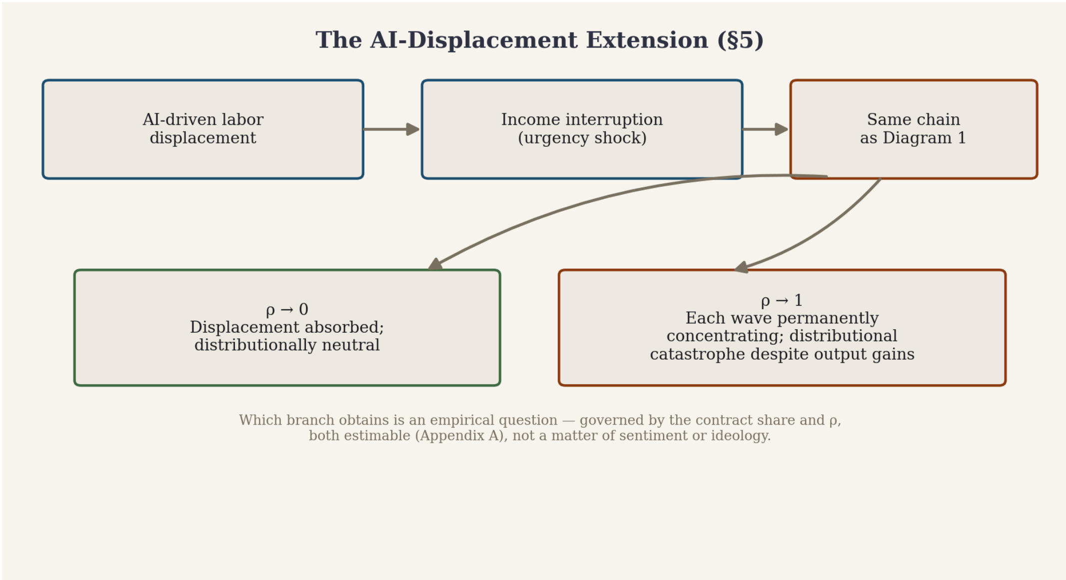

Figure 5. The AI-displacement extension. Which branch obtains is empirical — governed by the contract share and ρ, both estimable (Appendix A), not a matter of sentiment or ideology. [MODEL]

6. What follows — measurement and policy

Measurement. Estimate the contract share and ρ directly from balance-sheet and flow data, rather than reading a preference parameter off trade data where it is not identified [OPEN].

Policy. Bridge liquidity — lending against a solvent position to prevent a forced sale — is, in this model, not a bailout but a way to hold ρ down during a transition. That yields a concrete cost-benefit for bridge-liquidity facilities as macro-stabilization [MODEL].

Falsifiable prediction. Displacement should bind first where the contract share is highest — routine-cognitive, high debt-to-income cohorts (mid-level administrative, junior software, paralegal): highest AI-substitution exposure paired with highly leveraged balance sheets — a claim the data can refute [OPEN].

The through-line: order-dependence and concentration are carried by whatever forces amplitude to move — here, the contract that compels the sale — never by preference alone. Get the liquidity state into the model and the “panic” dissolves into something you can measure, and therefore something policy can act on.

References

Verified against source (DOIs valid, bibliographic details confirmed, July 2026). Each entry is annotated with the claim it supports and that claim's tag.

Shleifer, A., & Vishny, R. W. (2011). Fire Sales in Finance and Macroeconomics. Journal of Economic Perspectives, 25(1), 29–48. (NBER WP 16642, 2010.) — Anchors §1, the compulsion-vs-preference baseline [MODEL]: a fire sale is a forced sale at a dislocated price, definitionally independent of the seller's risk preference. Also motivates ρ < 1 — the natural buyers are themselves constrained, which is why discounts arise.

Carroll, C. D., & Samwick, A. A. (1997). The Nature of Precautionary Wealth. Journal of Monetary Economics, 40(1), 41–71. — Anchors the buffer-depletion lag [MODEL]: households hold a precautionary liquid buffer against near-term income shocks, so a shock depletes the buffer before it binds. Grounds why a lag exists; the lag length remains [DATA] to measure.

Kaplan, G., Violante, G. L., & Weidner, J. (2014). The Wealthy Hand-to-Mouth. Brookings Papers on Economic Activity, Spring, 77–153. (NBER WP 20073.) — Anchors the contract-share argument and cohort identification [DATA]: “wealthy hand-to-mouth” households hold sizable illiquid, often encumbered assets and near-zero liquid wealth, so an income interruption binds immediately.

Campbell, J. Y., Giglio, S., & Pathak, P. (2011). Forced Sales and House Prices. American Economic Review, 101(5), 2108–2131. — Anchors the price penalty of the urgency gap [DATA]: forced sales transact at large discounts (~27% for foreclosures) with local spillovers. NB: grounds the discount, not ρ — the absorption/holding coefficient is distinct (Appendix A).

Secondary anchors to verify before citing: Mian, Sufi & Trebbi (judicial vs. non-judicial foreclosure as an instrument); Mills, Molloy & Zarutskie (large-scale buy-to-rent investors). Not yet source-checked — not cited until verified.

Appendix A — Estimating the harvest coefficient ρ

ρ carries the whole magnitude of the result ($A = 1/(1-\rho)$; $\rho = 0.7 \Rightarrow A \approx 3.3$). It must be estimated, not assumed; it is currently [OPEN]. No citation in the reference list pins it down.

Setting. The 2008–2014 liquidation of distressed single-family homes (foreclosure / REO / short sale) is the cleanest available natural setting: forced supply, buyer identity, and subsequent holding are all observable in deed records.

Make ρ operational — decompose into two observables: ρ_{absorb} = share of distressed transactions bought by unconstrained buyers (all-cash / institutional LLC) rather than leveraged owner-occupants; $h(T)$ = of those, the share still held (converted to rental / not resold to an owner-occupant) at horizon T; $\rho(T) = \rho_{absorb} \times h(T)$ — the share of forced supply that reaches unconstrained hands and stays.

Data. Transaction/deed level: CoreLogic or ATTOM [paid]. Free proxies: HMDA (owner-occupant vs. investor flags — but misses all-cash buyers, i.e. precisely the unconstrained investors, a limitation to state up front); Census/ACS tenure; FHFA and the GSE REO-to-Rental pilot records.

Identification. Investor demand is endogenous to the discount (they enter where distress is deepest), biasing ρ_{absorb} upward. Instrument forced-supply pace with the judicial vs. non-judicial foreclosure regime, which shifts the pace for legal rather than demand reasons.

Report $\rho(T)$ across holding horizons — “permanent” is horizon-dependent.

Validation. If ρ is real, higher forced supply × unconstrained-buyer presence should produce larger, more persistent rises in investor-owned SFH share and declines in owner-occupancy, scaling with $1/(1-\rho)$. Placebo: non-distressed sales show no ratchet. Only after estimation does ρ move from [OPEN] to [DATA].

Appendix B — Formal verification (Lean 4)

The amplification result of §4 is the paper's one machine-checkable claim; the empirical claims (§§1, 3, and the value of ρ) are not theorems and are deliberately not formalized. Encoding an empirical claim as a trivially-true statement would misrepresent its status.

The supplementary Lean 4 file ForcedUrgency.lean certifies, against Mathlib: (i) the cascade ∑ ρ^k sums to $A(\rho) = 1/(1-\rho)$ for $0 \le \rho$ < 1; (ii) $A(\rho) \ge 1$; (iii) $A(\rho) = 1$ iff $\rho = 0$; (iv) A is strictly increasing in ρ; (v) A is unbounded as $\rho \to 1$ (filter-free form); (vi) policy corollary: any intervention that strictly lowers ρ strictly lowers cumulative displacement.

Kernel status: pending first run. Before any result here is labeled verified, #print axioms must show only [propext, Classical.choice, Quot.sound] — no sorryAx. The file's [OPEN] block lists what is not formalized: the identification failure, the buffer lag, and the estimated value of ρ. (Appendix C reserved.)

Appendix D — Policy application note

For use by policy organizations, think tanks, and advocacy groups evaluating whether the ρ / contract-share framework is applicable to their own data. External figures below are cited to their original sources; they have not been independently re-verified in this build and carry the tags assigned in the source discussion.

D.1 Purpose. This note lets an outside organization (i) check the framework's claims against its own primary sources, and (ii) reuse ρ and the contract share as measurement tools, without re-deriving them. Every empirical claim below is cited to its original source, not to this paper's paraphrase.

D.2 The wealth-concentration data. Oxfam International (2026), “Resisting the Rule of the Rich” (released 19 January 2026, Davos): global billionaire wealth rose over 16% in 2025 to a record, three times the prior five-year average, and 81% since 2020 [DATA — Oxfam 2026]. The richest 1% own nearly 45% of global wealth; 44% of humanity lives below the World Bank's $6.85/day line [DATA — Oxfam, citing World Bank]. Billionaires are ~4,000 times more likely to hold political office than ordinary citizens [DATA — Oxfam 2026, Oxfam's own estimate, to be cited as such]. Note for reuse: Oxfam's billionaire figures derive from the Forbes Real-Time Billionaire List — a market-value snapshot of a specific list, not a Z.1-style aggregate; the two data types are not directly comparable without adjustment and should not be combined in one regression.

D.3 The preventable-death data. WHO (2026), Child mortality (under 5): the global under-five mortality rate fell to 37.4 per 1,000 live births in 2024, less than half its 2000 level; more than 27 million children are projected to die before age five between 2025 and 2030, most from preventable causes [DATA — WHO 2026]. UNICEF (2025), State of the World's Children: more than 19% of children live in extreme monetary poverty (under $3/day) [DATA — UNICEF 2025]. UNICEF Data (2026): the annual rate of reduction in under-five mortality fell from 3.9%/yr (2000–2015) to 1.5%/yr (2015–2024). The Lancet (via UNICEF USA, 2026): aid cuts could cause 4.5 million additional under-five deaths by 2030 [cited via secondary source; primary Lancet article to be located before formal use — OPEN, not a complete citation].

D.4 What the framework adds. Oxfam documents that concentration is happening and quantifies its scale; it does not provide a mechanism-level coefficient for why a given shock is diffused versus captured. That is the contribution of ρ and the contract share: a parameter, estimable from data (Appendix A), that converts “concentration is increasing” into a mechanism with a predicted magnitude, $A = 1/(1-\rho)$, checkable against a specific policy change before and after. This is complementary to Oxfam's advocacy data, not a replacement or critique.

D.5 Methodology transparency required before external reuse. Because ρ determines how a policy question gets answered, whoever controls its inputs has real influence over the conclusion. Following the same disclosure standard as this author's Lean proof audits (open axiom lists): all data and code used to estimate ρ_{absorb} and $h(T)$ should be published openly so any organization can rerun the estimate; no single ρ estimate is final (report $\rho(T)$ across horizons, state the endogeneity instrument); any organization should be able to substitute its own data source and get a comparable, checkable output. This note should be updated if a more current or precisely sourced figure than those in D.2–D.3 is identified.

References (Appendix D): Oxfam International (2026), Resisting the Rule of the Rich; Oxfam International (2025), Takers Not Makers; WHO (2026), Child mortality (under 5 years);

UNICEF (2025), State of the World's Children; UNICEF Data (2026), Under-five mortality; The Lancet (via UNICEF USA 2025 summary) [OPEN — primary citation to confirm]; World Bank (2025), June 2025 poverty-line update.